



# THE SCOTS COLLEGE

2003  
TRIAL HSC EXAMINATION

## MATHEMATICS EXTENSION 1

### GENERAL INSTRUCTIONS

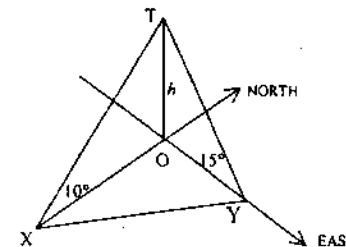
- Reading time - 5 minutes
- Working time - 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- A table of integrals is provided
- All necessary working should be shown
- Start each question on a new booklet
- Attempt Questions 1 - 7
- All questions are of equal value

### QUESTION 1

- (a) Find the acute angle between the lines  $2x - y = 0$  and  $x + 3y = 0$ , giving the answer correct to the nearest minute. [1]
- (b) Solve the inequality  $\frac{x}{x-3} \leq 3$  [3]
- (c) If  $u, v$  and  $w$  are the roots of  $x^3 - 4x + 1 = 0$ , find the value of  $\frac{1}{u} + \frac{1}{v} + \frac{1}{w}$ . [3]
- (d) Solve the equation  $\sin 2x = \tan x$  for  $0 \leq x \leq \pi$ . [4]

### QUESTION 2 [START A NEW BOOKLET]

- (a) A is the point  $(-2, 1)$  and B is the point  $(x, y)$ . The point  $P(13, -9)$  divides AB externally in the ratio  $5 : 3$ . Find the values of  $x$  and  $y$ . [3]
- (b) (i) Show that the equation of the normal to the parabola  $x^2 = 4ay$  at the point  $T(2at, at^2)$  is  $x + ty = 2at + at^3$ . [2]
- (ii) Hence show that there is only one normal to the parabola which passes through its focus. [1]
- (c) A surveyor at X observes a tower due north. The angle of elevation to the top of the tower is  $10^\circ$ . He then walks 400m to a position Y which is due east of the tower. The angle of elevation from Y to the top of the tower is  $15^\circ$ .

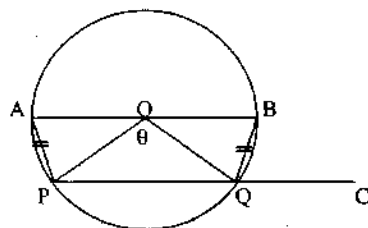


- (i) Write an expression for OY in terms of  $h$ . [1]
- (ii) Calculate  $h$  to the nearest metre. [4]
- (iii) Find the bearing of Y from X. [1]

QUESTION 3 [START A NEW BOOKLET]

(a) Evaluate  $\int_0^{2\pi} \cos^2 2x \, dx$ . [3]

(b)



The points A, B, P and Q lie on the circle with centre at O.

AB is a diameter and PC passes through Q.

AP is equal to BQ and  $\angle POQ = \theta$

(i) Express  $\angle AOP$  in terms of  $\theta$ . [1]

(ii) Prove that AB is parallel to PC. [2]

(c) By graphing or some other justification, simplify [3]

(i)  $\sin^{-1} x + \sin^{-1}(-x)$

(ii)  $\tan^{-1} x + \tan^{-1}(-x)$

(iii)  $\sin^{-1} x - \cos^{-1}(-x)$

(d) Find  $\int_0^2 2x \sqrt{1 - \frac{x}{2}} \, dx$  using the substitution  $u = 1 - \frac{x}{2}$  [3]

QUESTION 4 [START A NEW BOOKLET]

(a) The surface area of a cube is increasing at a rate of  $10\text{cm}^2$  per second. Find the rate of increase of the volume of the cube when the edge of the cube has length 12cm. [4]

(b)  $N$  is the number of animals in a certain population at time  $t$  years. The population size  $N$  satisfies the equation  $\frac{dN}{dt} = -k(N - 1000)$  for some constant  $k$ .

(i) Verify where  $A$  is constant, that  $N = 1000 + Ae^{-kt}$  is a solution of the equation. [2]

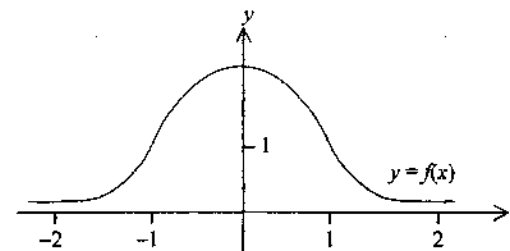
(ii) Initially there are 2500 animals but after 2 years there are only 2200 left. Find the values of  $A$  and  $k$ , to 2 decimal places. [2]

(iii) Find when the number of animals has fallen to 1300. [2]

(iv) Sketch the graph of the population size against time. [2]

QUESTION 5 [START A NEW BOOKLET]

(a) The graph below shows the derivative of  $y = 2 \tan^{-1} x$ .



(i) Where does  $y = 2 \tan^{-1} x$  have its greatest slope and what is this slope? [2]

(ii) Calculate the  $x$  values correspond with  $\frac{dy}{dx} = \frac{1}{3}$ ? [1]

(iii) Write an integral that represents the area in the first quadrant bounded by this curve, the  $x$  axis and  $x = k$ , where  $k > 0$ . [1]

(iv) By considering the limit as  $k \rightarrow \infty$  determine the total area bounded by this curve and the  $x$  axis. [1]

(b) (i) Sketch the graph of function  $f(x) = e^x - 4$ . [1]

(ii) On the same diagram sketch the graph of the inverse function  $f^{-1}$ . [2]

(iii) Explain why the  $x$  coordinate of any point of intersection of the graphs  $y = f(x)$  and  $y = f^{-1}(x)$  satisfies the equation  $e^x - x - 4 = 0$ . [1]

(iv) Show that the equation  $e^x - x - 4 = 0$  has a root between  $x = 1$  and  $x = 2$ . Use one application of Newton's method to approximate the root, to 2 decimal places. [3]

QUESTION 6 [START A NEW BOOKLET]

(a) Prove by Mathematical Induction that  $1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$  for all positive integers  $n$ . [5]

(b) A particle moves in a straight line so that its displacement  $x$  from a fixed point  $O$  at time  $t$  is given by  $x = 3 \sin 2t + 4 \cos 2t$ .

(i) If the motion is expressed in the form of  $x = R \sin(2t + \alpha)$  where  $\alpha$  is in radians, evaluate the constants  $R$  and  $\alpha$ , to 2 decimal places. [3]

(ii) Show that the motion is Simple Harmonic. [1]

(iii) What is the period of oscillation? [1]

(iv) Determine the maximum displacement from the centre of motion. [2]

QUESTION 7 [START A NEW BOOKLET]

(a) A projectile has an initial velocity  $V$  and an angle of projection  $\theta$ .

(i) Assuming  $\frac{d^2y}{dt^2} = -10$ ,  $\frac{d^2x}{dt^2} = 0$  and initially  $x = 0$ ,  $y = 10$ , find expressions for  $x$  and  $y$ .

[3]

(ii) If  $V = 13 \text{ ms}^{-1}$  and  $\theta = \tan^{-1}\left(\frac{5}{12}\right)$  find the range of the projectile.

[2]

(b) (i) Use the Chain Rule to show that

$$\frac{dv}{dt} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

[1]

(ii) The acceleration due to gravity is inversely proportional to the square of the distance  $x$  from the centre of the earth.

This can be written as  $\frac{dv}{dt} = \frac{-k}{x^2}$ . Find  $k$  if  $\frac{dv}{dt} = -g$  when  $x = R$ .

[1]

(iii) Hence show that  $v^2 = \frac{2R^2g}{x} + u^2 - 2gR$  where the initial velocity of a rocket is  $u \text{ ms}^{-1}$ ,  $g$  is the acceleration due to gravity and  $R$  is the radius of the earth.

[2]

(iv) Find the maximum distance that the rocket will travel from the centre of the earth. (Answer in terms of  $g$ ,  $R$  and  $u$ ).

[2]

(v) Taking  $g = 9.8 \text{ ms}^{-2}$ ,  $R = 6400 \text{ km}$  find the value of  $u$  in  $\text{ms}^{-1}$  for which the rocket will escape the gravity of the earth.

[1]

Question 1

a)  $m_1 = 2, m_2 = -\frac{1}{3}$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 + \frac{1}{3}}{1 - 2(\frac{1}{3})} \right|$$

$$= |7|$$

$$\therefore \theta = 81^\circ 52'$$

b)  $\frac{x}{x-3} \leq 3$

$$x(x-3) \leq 3(x-3)^2$$

$$x^2 - 3x \leq 3x^2 - 18x + 27$$

$$2x^2 - 15x + 27 \geq 0$$

$$(2x-9)(x-3) \geq 0$$

$$\therefore x < 3 \text{ or } x \geq \frac{9}{2}$$

c)  $x^3 - 4x + 1 = 0$

$$u + v + w = 0$$

$$uv + uw + vw = -4$$

$$uvw = -1$$

$$\frac{1}{u} + \frac{1}{v} + \frac{1}{w} = \frac{vw + uw + uv}{uvw}$$

$$= \frac{-4}{-1}$$

$$= 4$$

d)  $\sin 2x = \tan x \quad 0 \leq x < \pi$

$$2 \sin x \cos x = \frac{\sin x}{\cos x}$$

$$2 \sin x \cos^2 x - \sin x x = 0$$

$$\sin x (2 \cos^2 x - 1) = 0$$

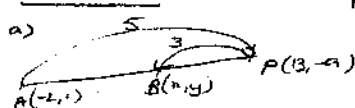
$$\sin x = 0 \text{ or } \cos x = \pm \frac{1}{\sqrt{2}}$$

$$x = 0, \pi$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

Question 2



$$x_1 = \frac{nx_1 + mx_2}{m+n} \quad y = \frac{ny_1 + my_2}{m+n}$$

$$13 = \frac{-3(-2) + 5x}{5-3} \quad -9 = \frac{-3(1) + 5y}{5-3}$$

$$26 = 6 + 5x$$

$$5x = 20$$

$$x = 4$$

$$-18 = -3 + 5y$$

$$5y = -15$$

$$y = -3$$

b)  $x^2 = 4ay$

(i)  $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{x}{2a}$$

At  $x = 2at$ ,

$$\frac{dy}{dx} = t$$

(ii)  $S(0,a)$

$$at = 2at + at^3$$

$$at + at^3 = 0$$

$$at(1 + t^2) = 0$$

$$\therefore t = 0 \text{ or } t^2 = -1 \text{ no solution.}$$

c) (i)  $\tan 15^\circ = \frac{h}{OY}$

$$OY = \frac{h}{\tan 15^\circ}$$

(ii)  $\tan 10^\circ = \frac{h}{OX}$

$$OX = \frac{h}{\tan 10^\circ}$$

$$OX^2 + OY^2 = 400^2$$

$$\frac{h^2}{\tan^2 10^\circ} + \frac{h^2}{\tan^2 15^\circ} = 160000$$

$$h^2 \left( \frac{1}{\tan^2 10^\circ} + \frac{1}{\tan^2 15^\circ} \right) = 160000$$

$$h^2 = 3471.345$$

$$h = 58.918$$

$$h = 59 \text{ m}$$

(iii)  $\tan \theta = \frac{OY}{400}$

$$\sin \theta = \frac{59}{400 \tan 15^\circ}$$

$$\theta = 0.33^\circ \text{ T}$$

Question 3

a)  $\cos^2 x = \frac{1}{2} (2 \cos 2x + 1)$

$$\cos^2 2x = \frac{1}{2} (2 \cos 4x + 1)$$

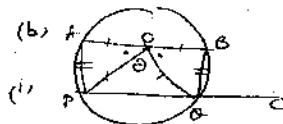
$$\int_0^{2\pi} \cos^2 2x \, dx$$

$$= \frac{1}{2} \int_0^{2\pi} (2 \cos 4x + 1) \, dx$$

$$= \frac{1}{2} \left[ \frac{\sin 4x}{4} + x \right]_0^{2\pi}$$

$$= \frac{1}{2} \left( \frac{\sin 8\pi}{4} + 2\pi - \left( \frac{\sin 0}{4} + 0 \right) \right)$$

$$= \pi$$



$$\triangle AOP \equiv \triangle BOQ \text{ (SSS)}$$

$$\therefore \angle AOP = \angle BOQ$$

$$\therefore \angle AOP = \frac{180 - \theta}{2}$$

$$= 90^\circ - \frac{\theta}{2}$$

(ii)  $\angle OPA = \frac{180 - \theta}{2}$  (base  $\angle$ s of isosceles  $\triangle$ )

$$= 90^\circ - \frac{\theta}{2}$$

$$\therefore \angle AOP = \angle OPA \text{ and they are alternate.}$$

$$\therefore AB \parallel PO$$

c) (i)  $\sin^{-1}(x) + \sin^{-1}(-x)$

$$= \sin^{-1}(x) - \sin^{-1}(x)$$

$$= 0$$

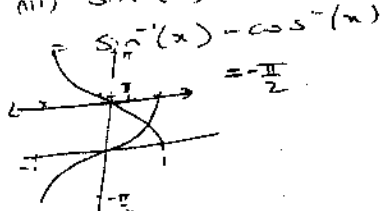
(ii)  $\tan^{-1}(x) + \tan^{-1}(-x)$

$$= \tan^{-1}(x) - \tan^{-1}(x)$$

$$= 0$$

(iii)  $\sin^{-1}(x) - \cos^{-1}(-x)$

$$= \sin^{-1}(x) - \cos^{-1}(x)$$



$$d) u = 1 - \frac{x}{2} \quad x=0 \quad u=1$$

$$x=2, u=0$$

$$\frac{du}{dx} = -\frac{1}{2} \quad x = 2(1-u)$$

$$\int_0^2 2x \sqrt{1 - \frac{x}{2}} dx$$

$$= 2 \cdot 2 \int_1^0 2(1-u) \sqrt{u} \cdot du$$

$$= -8 \int_0^1 u^{\frac{1}{2}} (1-u) du$$

$$= 8 \int_0^1 u^{\frac{1}{2}} - u^{\frac{3}{2}} du$$

$$= 8 \left[ \frac{2u^{\frac{3}{2}}}{3} - \frac{2u^{\frac{5}{2}}}{5} \right]_0^1$$

$$= 8 \left( \frac{2}{3} - \frac{2}{5} - 0 \right)$$

$$= \frac{32}{15}$$

#### Question 4

$$a) \frac{dA}{dt} = 10 \quad x=12$$

$$\frac{dV}{dt} = ?$$

$$A = 6x^2 \quad V = x^3$$

$$\frac{dA}{dt} = \frac{dA}{dx} \cdot \frac{dx}{dt}$$

$$10 = 12x \cdot \frac{dx}{dt}$$

$$x=12, 10 = 144 \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{10}{144}$$

$$\frac{dx}{dt} = \frac{5}{72}$$

$$\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$$

$$= 3x^2 \cdot \frac{dx}{dt}$$

$$= 3(12)^2 \cdot \frac{5}{72}$$

$$= 30 \text{ cm}^3/\text{s}$$

(b)

$$(i) N = 1000 + Ae^{-kt}$$

$$\frac{dN}{dt} = Ae^{-kt} \cdot -k$$

$$= -k(N - 1000)$$

$$(ii) t=0, N=2500$$

$$t=2, N=2200$$

$$2500 = 1000 + A$$

$$A = 1500$$

$$2200 = 1000 + 1500e^{-2k}$$

$$1500e^{-2k} = 1100$$

$$e^{-2k} = \frac{11}{15}$$

$$-2k = \ln\left(\frac{11}{15}\right)$$

$$k = \frac{\ln\left(\frac{11}{15}\right)}{-2}$$

$$k = 0.16 \text{ (2dp)}$$

$$(iii) 1300 = 1000 + 1500e^{-0.16t}$$

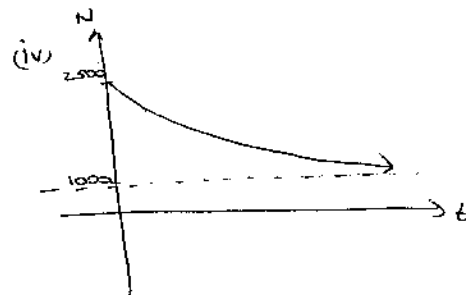
$$1500e^{-0.16t} = 300$$

$$e^{-0.16t} = \frac{1}{5}$$

$$-0.16t = \ln\left(\frac{1}{5}\right)$$

$$t = \frac{\ln\left(\frac{1}{5}\right)}{-0.16}$$

$$t = 10.06 \text{ years}$$



#### Question 5

$$a) i) y = 2 \tan^{-1} x$$

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

$$\text{greatest slope occurs at}$$

$$x=0, \frac{dy}{dx} = 2$$

$$(ii) \frac{2}{1+x^2} = \frac{1}{3}$$

$$6 = x^2 + 1$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$$(iii) A = \int_0^k f(x) dx$$

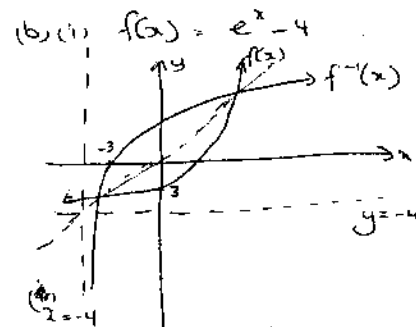
$$(iv) \text{As } k \rightarrow \infty, \text{ Area under curve}$$

$$A = 2 \int_0^{\infty} f(x) dx$$

$$= 2 \cdot \left[ 2 \tan^{-1} x \right]_0^{\infty}$$

$$= 2 \cdot 2 \cdot \frac{\pi}{2}$$

$$= 2\pi \text{ sq units.}$$



(ii)  $f(x)$  and  $f^{-1}(x)$  are reflections along the line  $y = x$ .

Points of intersection are

$$y = e^x - 4 \text{ and } y = x \text{ hold true}$$

$$\text{ie } e^x - 4 = x$$

$$e^x - x - 4 = 0.$$

$$(iv) \text{ let } f(x) = e^x - x - 4$$

$$f(1) = e - 1 - 4 < 0$$

$$f(2) = e^2 - 2 - 4 > 0$$

$\therefore$  root lies between  $x=1$  and  $x=2$

$$f'(x) = e^x - 1 \quad x_1 = 1.5$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.5 - \frac{f(1.5)}{f'(1.5)}$$

$$= 1.79 \text{ (2 dp)}$$

#### Question 6

a) Step 1: Need to prove  $n=1$  is true

$$\text{LHS} = 1, \quad \text{RHS} = \frac{1(3(1)-1)}{2}$$

$$= 1$$

$$= \text{LHS}$$

$\therefore n=1$  is true

Step 2: Assume that  $n=k$  is true

$$\text{ie } 1+4+7+\dots+(3k-2) = \frac{k(3k-1)}{2}$$

Need to prove that  $n=k+1$  is true

$$\text{ie } 1+4+7+\dots+(3k-2)+(3k+1)$$

$$= \frac{(k+1)(3k+2)}{2}$$

$$\text{LHS} = 1+4+7+\dots+(3k-2)+(3k+1)$$

$$= \frac{k(3k-1)}{2} + (3k+1)$$

$$= \frac{1}{2}(3k^2 - k + 6k + 2)$$

$$= \frac{1}{2} (3k^2 + 6k - 2)$$

$$= \frac{1}{2} (3k+1)(k-2)$$

$$= \text{RHS}$$

$\therefore n=k+1$  is also true

Step 3. Since,  $n=1$ ,  $n=k$  &  $n=k+1$  are all true

then  $n=2, n=3, \dots$  are true

$$1+4+7+\dots+(3n-2) = \frac{n(3n-1)}{2}$$

(b)  $x = 3\sin 2t + 4\cos 2t$

(i)  $3\sin 2t + 4\cos 2t$   
 $= R \sin(2t + \alpha)$   
 $= R \sin 2t \cos \alpha + R \cos 2t \sin \alpha$

$$\therefore R \cos \alpha = 3 \quad R^2 = 3^2 + 4^2$$

$$R \sin \alpha = 4 \quad R = \sqrt{25}$$

$$\tan \alpha = \frac{4}{3} \quad R = 5$$

$$\alpha = 0.93$$

(iii)  $x = 5 \sin(2t + 0.93)$   
 $\dot{x} = 10 \cos(2t + 0.93)$   
 $\ddot{x} = -20 \sin(2t + 0.93)$   
 $= -4x$

$\therefore$  motion is S.H.

(iii) period  $= \frac{2\pi}{2}$   
 $= \pi$

(iv) max deep when  $\dot{x} = 0$   
 $10 \cos(2t + 0.93) = 0$   
 $\cos(2t + 0.93) = 0$   
 $2t + 0.93 = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$   
 $2t = \frac{\pi}{2} - 0.93, \frac{3\pi}{2} - 0.93$   
 $t = 0.32, 1.89, \dots$

At  $t = 0.32$ ,  
 $x = 5 \sin(2(0.32) + 0.93)$   
 $x = 5$

Question 7

a)  $\frac{dx}{dt} = 0$  Initially,  
 $\dot{x} = v \cos \theta$   
 $\dot{y} = v \sin \theta$   
 $\frac{dx}{dt} = C_1$

$t=0, \frac{dx}{dt} = v \cos \theta, \therefore \frac{dx}{dt} = v \cos \theta$

$$x = \int v \cos \theta dt$$

$$x = vt \cos \theta + C_2$$

$t=0, x=0, C_2=0, \therefore x = vt \cos \theta$

$$\frac{d^2y}{dt^2} = -10$$

$$\frac{dy}{dt} = \int -10 dt$$

$$= -10t + C_3$$

$t=0, \frac{dy}{dt} = v \sin \theta, \therefore C_3 = v \sin \theta$

$$\therefore \frac{dy}{dt} = -10t + v \sin \theta$$

$$y = \int -10t + v \sin \theta dt$$

$$y = -5t^2 + vt \sin \theta + C_4$$

$t=0, y=10, \therefore C_4 = 10$

$$\therefore y = -5t^2 + vt \sin \theta + 10$$

(i)  $v=13$  

Sub  $y=0, -5t^2 + t \cdot 13 \cdot \frac{5}{13} + 10 = 0$

$$-5t^2 + 5t + 10 = 0$$

$$t^2 + t - 2 = 0$$

$$(t-2)(t+1) = 0$$

$$\therefore t = 2 \text{ or } t = -1 \text{ but } t \geq 0$$

$$\therefore t = 2$$

When  $t = 2, x = 13 \cdot 2 \cdot \frac{12}{13} = 24m$

b) (i)  $\frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt}$   
 $= v \cdot \frac{dv}{dx}$

$$\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = \frac{1}{2} \cdot 2v \cdot \frac{dv}{dx}$$

$$= v \cdot \frac{dv}{dx}$$

$$\therefore \frac{dv}{dt} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

(iii)  $\frac{dv}{dt} = -\frac{k}{x^2}$

Sub  $\frac{dv}{dt} = -g, x = R,$

$$-g = -\frac{k}{R^2}$$

$$R^2 g = k$$

(iii)  $\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = -\frac{R^2 g}{x^2}$

$$\therefore \frac{1}{2} v^2 = \int \frac{R^2 g}{x^2} dx$$

$$\frac{1}{2} v^2 = \frac{R^2 g}{x} + C$$

When  $x = R, v = u$

$$\frac{1}{2} u^2 = \frac{R^2 g}{R} + C$$

$$\therefore C = \frac{1}{2} u^2 - Rg$$

$$\therefore \frac{1}{2} v^2 = \frac{R^2 g}{x} + \frac{1}{2} u^2 - Rg$$

$$\therefore v^2 = \frac{2R^2 g}{x} + u^2 - 2gR$$

(iv) max distance,  $v=0$

$$\frac{2R^2 g}{x} + u^2 - 2gR = 0$$

$$\frac{2R^2 g}{x} = 2gR - u^2$$

$$\therefore x = \frac{2R^2 g}{2gR - u^2}$$

(vi)  $g = 9.8, R = 6400$

as  $x \rightarrow \infty, u^2 = 2gR$

$$u^2 = 2(9.8)(6400)$$

$$u = \pm 11200$$

but  $u > 0 \therefore u = 11200 \text{ m s}^{-1}$